

Changelog

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This document records the changes made to Deliverable 1 *after* its submitted version. The revisions to that submission, made in response to the feedback of 4 June 2026, are recorded in the changelog that accompanied it and are not repeated here. Only one file changed: the body section. The equation count rises from ten to thirteen, so the contrast is now Eq. (8) rather than Eq. (9) while the Gamma regression remains Eq. (10).

Operational definition of structural aliasing

The submitted version defined structural aliasing by a representation-distance threshold: two frequency-matched signals were said to alias at layer l when $\|R_l(x) - R_l(y)\|_2 \leq \varepsilon$ for a pre-registered ε . That equation has been removed. The project’s own probing does not support it, at a lock the token is a geometric *outlier* rather than a near-collision, so the criterion would have committed the report to a claim the measurements contradict. Structural aliasing is now defined by readability: the phenomenon is present when the signal’s frequency can no longer be recovered from the model’s internal state at a locked frequency as well as at

Change	What was done	Where
Operational definition of structural aliasing	The representation-distance criterion $\ R_l(x) - R_l(y)\ _2 \leq \varepsilon$ was withdrawn and replaced by a readability criterion, since the probing evidence contradicts the near-collision reading.	Mathematical Formulation
Bayesian analysis extended to H2 and H3	One model per hypothesis. The H1 framework is unchanged; a phase model and two site-geometry models were added, with priors and decision rules.	Bayesian Analysis, Eqs. (11)–(13)
Local-contrast hierarchy stated	The grouping implied by μ_i is written out: configuration, harmonic and background levels, plus an overlap slope δ_O .	Bayesian Analysis, Eq. (9)
Codelength response defined	The Gamma response is specified as a frequency-local codelength, so that the IsLocked indicator has something to attach to.	Bayesian Analysis, Eq. (10)
Inference protocol stated	Sampler settings, acceptance thresholds and the validation sequence replace the former one-line statement.	Bayesian Analysis

Table 1: *Changes from the submitted Deliverable 1 to this revision.*

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During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.6 Flash, Sonnet 5, and Opus 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

neighbouring, non-locked ones. This is the criterion the experiments actually test, and it is what the *H1* metrics already measure.

Bayesian analysis: extension to H2 and H3

The submitted version applied its two-part framework to *H1* alone; the lecturers asked for the treatment to be extended. Each hypothesis now has its own estimand and model, and the *H1* models are unchanged in substance.

Two additions are new. *Phase-Invariance Analysis (H2)* retains the phase of each measurement, which the contrast of Eq. (8) averages away, cuts the phase circle into eight bins and gives each bin its own offset (Eq. 11); the spread σ_ϕ of those offsets is tested against a preregistered region of practical equivalence and against the phase-free model by leave-one-out. *Site-Geometry Analysis (H3)* targets the location of the effect rather than its size: the token-collapse profile is regressed on indicators of the two predicted grids (Eq. 12), swept on a grid that unions every configuration's predicted sites so no geometry is scored on a grid of its own, with the stride-only, patch-only and null variants compared by leave-one-out; a second model regresses the measured comb spacing on the predicted spacings f_s/S and f_s/P (Eq. 13), testing the slopes $\kappa_S = 1$ and $\kappa_P = 0$. Because the design varies P only along $P = S$, κ_P is declared partially confounded and the fixed- $P=16$ series carries the identification.

Both models were chosen for defensibility as much as for fit: a variance component and a linear regression on predicted values assume less, and are easier to state, than the first-harmonic and log-log formulations they replaced.

Local-contrast hierarchy and codelength response

Two specifications that the submitted version left implicit are now written out. The contrast likelihood is given in hierarchical form (Eq. 9), with configuration, harmonic and background levels and an overlap slope δ_O , and the inference is declared conditional on the single available training seed.

The Gamma regression keeps its form and priors, but its response is now defined. The seven band tasks span the whole operational band, so a single codelength per probe point summarises hundreds of frequencies and leaves the IsLocked indicator of Eq. (10) nothing to attach to. The response is therefore a *frequency-local* codelength: at each candidate frequency a balanced probe separates tones at $f_c \pm 1$ Hz from the same [REG] representation under an otherwise unchanged prequential protocol. The band-task values are retained as a descriptive cross-check.

Inference

The former single sentence is replaced by the full protocol: the quantities reported for each model, the sampler settings and the acceptance thresholds ($\hat{R} < 1.01$, effective sample sizes above 1000, no divergent transitions), and the validation sequence, prior predictive checks before data collection, simulation-based parameter recovery, stratified posterior predictive checks, and sensitivity over the prior-scale ladder.